

HW06: R Functions HW

Seunghoon Oh (PID: A19372132)

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Section 1B

The original code

I revised the typo in the line for s3.chainA (s1 to s3)

```
# Can you improve this analysis code?
library(bio3d)
s1 <- read.pdb("4AKE") # kinase with drug
```

Note: Accessing on-line PDB file

```
s2 <- read.pdb("1AKE") # kinase no drug
```

Note: Accessing on-line PDB file

PDB has ALT records, taking A only, rm.alt=TRUE

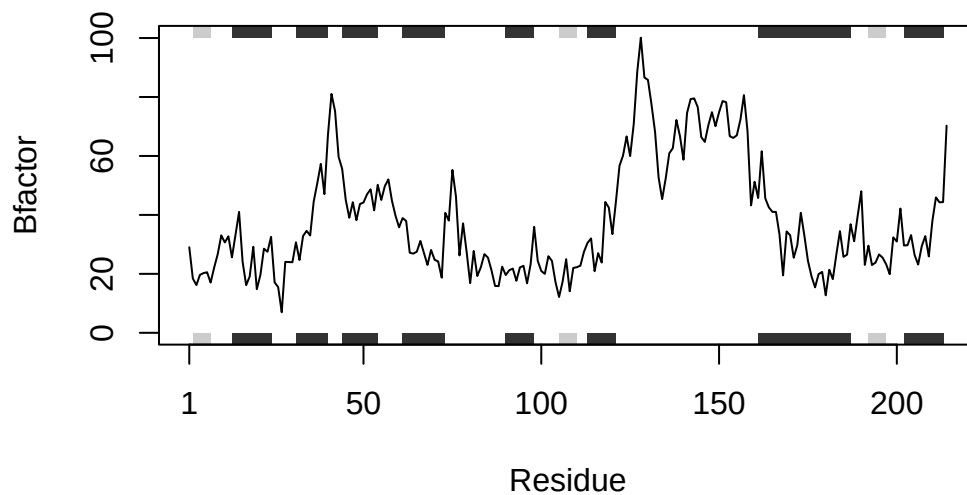
```
s3 <- read.pdb("1E4Y") # kinase with drug
```

Note: Accessing on-line PDB file

```

s1.chainA <- trim.pdb(s1, chain="A", elety="CA")
s2.chainA <- trim.pdb(s2, chain="A", elety="CA")
s3.chainA <- trim.pdb(s3, chain="A", elety="CA")
s1.b <- s1.chainA$atom$b
s2.b <- s2.chainA$atom$b
s3.b <- s3.chainA$atom$b
plotb3(s1.b, sse=s1.chainA, typ="l", ylab="Bfactor")

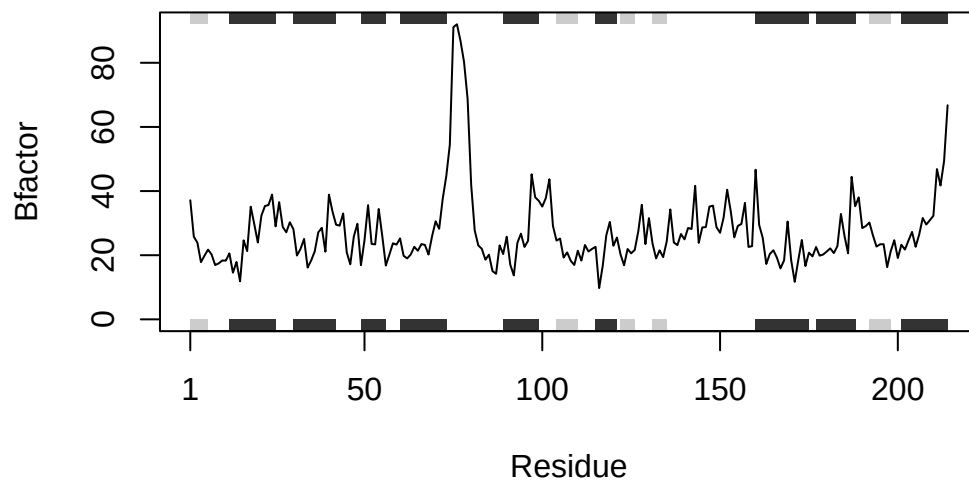
```



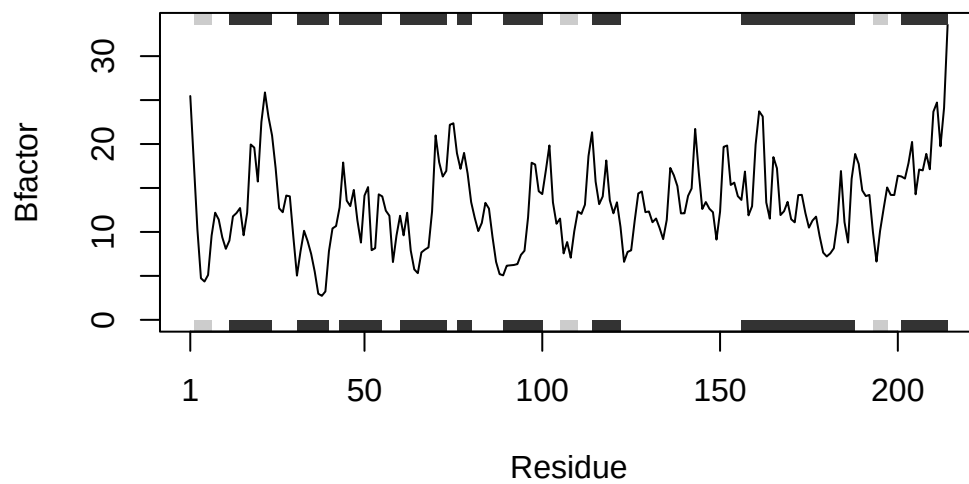
```

plotb3(s2.b, sse=s2.chainA, typ="l", ylab="Bfactor")

```

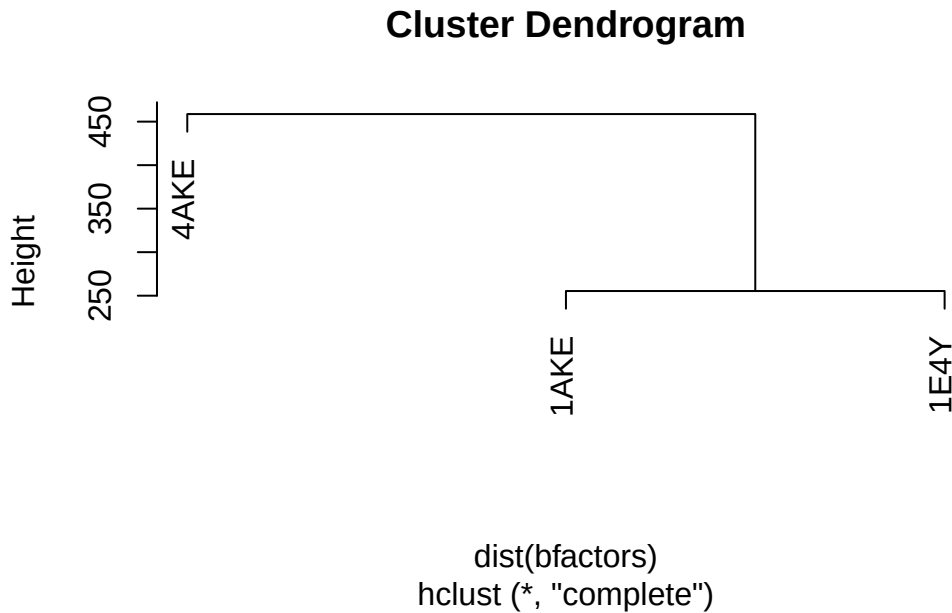


```
plotb3(s3.b, sse=s3.chainA, typ="l", ylab="Bfactor")
```



Q5

```
bfactors <- rbind(s1.b, s2.b, s3.b)
rownames(bfactors) <- c("4AKE", "1AKE", "1E4Y")
hc <- hclust(dist(bfactors))
plot(hc)
```



Q6

The original code repeats an exact same process (code) for all three pdb files 4AKE, 1AKE and 1E4Y: reading a PDB file, trimming chain A to CA atoms, extracting the B-factors, and plotting them in a same format. By defining a function `analyze_pdb_bfactor()`, we can generalize this for any input pdb files.

There are three **inputs** for the function: **pdb_id**, which is a PDB ID or file name, e.g. “4AKE”; **chain**, which is the protein chain we are analyzing; **elety**, the atom type we keep while trimming the pdb input. I have set the default value for **chain** as “A” and **elety** as “CA”, as set in the original code.

Following **the process** of the original code, this function reads a PDB file with structure data, trims it to the selected chain and atom type, extracts the B-factor value, and plots the B-factor

trend across the protein. The **output** is a B-factor plot for the selected protein structure. It also returns the B-factor values for future reuse.

```
analyze_pdb_bfactor <- function(pdb_id, chain = "A", elety = "CA") {  
  pdb <- read.pdb(pdb_id)  
  pdb_trimmed <- trim.pdb(pdb, chain = chain, elety = elety)  
  b_factors <- pdb_trimmed$atom$b  
  
  plotb3(  
    b_factors,  
    sse = pdb_trimmed,  
    typ = "l",  
    ylab = "B-factor",  
    main = paste("B-factor plot for", pdb_id)  
  )  
  
  return(b_factors)  
}
```

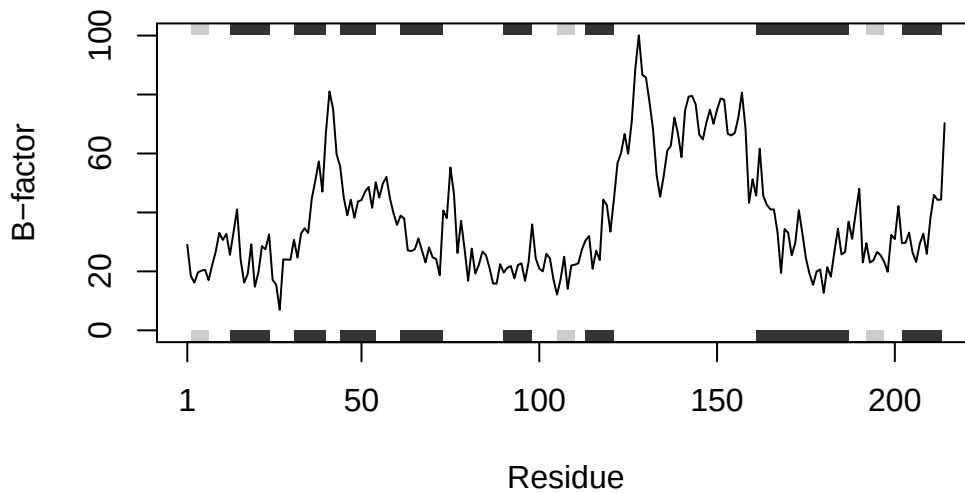
And we can apply the function for s1-s3 data.

```
s1.b <- analyze_pdb_bfactor("4AKE")
```

Note: Accessing on-line PDB file

```
Warning in get.pdb(file, path = tempdir(), verbose = FALSE):  
/var/folders/56/bn1513ts6_q_x93tzb1_q33r0000gn/T//RtmpTLTscX/4AKE.pdb exists.  
Skipping download
```

B-factor plot for 4AKE



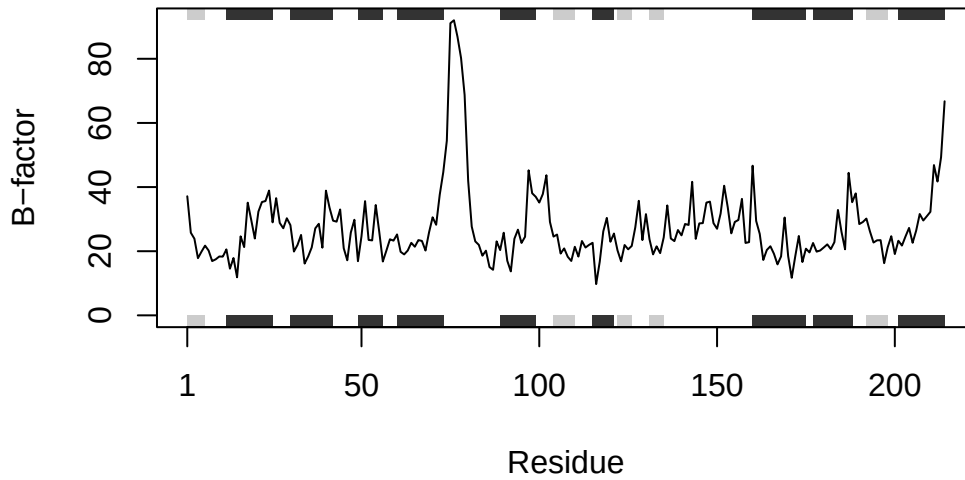
```
s2.b <- analyze_pdb_bfactor("1AKE")
```

Note: Accessing on-line PDB file

Warning in get.pdb(file, path = tempdir(), verbose = FALSE):
/var/folders/56/bn1513ts6_q_x93tzb1_q33r0000gn/T/RtmpTLTscX/1AKE.pdb exists.
Skipping download

PDB has ALT records, taking A only, rm.alt=TRUE

B-factor plot for 1AKE



```
s3.b <- analyze_pdb_bfactor("1E4Y")
```

Note: Accessing on-line PDB file

Warning in get.pdb(file, path = tempdir(), verbose = FALSE):
/var/folders/56/bn1513ts6_q_x93tzb1_q33r0000gn/T/RtmpTLTscX/1E4Y.pdb exists.
Skipping download

B-factor plot for 1E4Y

